Luiz Felipe Vecchietti

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Research Interests

My research aims to develop artificial intelligence algorithms that advance our knowledge of how humans and machines learn, interact, and collaborate. In particular, I am interested in reinforcement learning (credit assignment, goal-conditioned RL, continual learning), multi-agent systems (collaborative, collaborative-competitive), robotics (end-to-end control, foundation models, and safety risks), and AI for science (generative AI for protein design, flexible AI architectures). My research is being applied to diverse domains such as robotics, computational biology, demographic research, natural language processing, and speech processing.

Academic Positions

Max Planck Institute for Security and Privacy (MPI-SP) *Bochum, Germany, Nov. 2024 - Present*

Postdoctoral Researcher - Data Science for Humanity Group. Advisor: Meeyoung Cha.

Institute for Basic Science (IBS) *Daejeon, South Korea, Oct. 2021 - Sep. 2024*

Postdoctoral Researcher - Data Science Group. Advisor: Meeyoung Cha.

Mechanical Engineering Research Institute, KAIST *Daejeon, South Korea, Mar. 2021 - Sep. 2021*

Postdoctoral Researcher.

Education

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea, Aug. 2017 - Feb. 2021*

Ph.D. in Green Transportation. Advisor: Dongsoo Har.

Thesis topic: Performance Enhancement in Multigoal Reinforcement Learning using Hindsight Experience Replay.

COPPE - Federal University of Rio de Janeiro (UFRJ) *Rio de Janeiro, Brazil, Mar. 2015 - Apr. 2017*

M.Sc. in Electrical Engineering. Advisor: Fernando Gil Vianna Resende Junior.

Thesis topic: Comparison between rule-based and data-driven Natural Language Processing algorithms for Brazilian Portuguese Speech Synthesis.

Federal University of Rio de Janeiro (UFRJ) *Rio de Janeiro, Brazil, Feb. 2009 - Dec. 2014*

B.Sc. in Electronics and Computer Engineering.

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea, Feb. 2013 - Feb. 2014*

Exchange Student in Electrical Engineering (Science without Borders Program).

Publications

[1] M. S. Locatelli, F. Tonucci, J. Kwon, **L. F. Vecchietti**, B. N. Wijaya, C. Y. Low, V. Almeida, M. Cha, "**Textual Supervision Enhances Geospatial Representations in Vision-Language Models**," in International Conference on Machine Learning (ICML), 2026.

[2] N. Koeppe, **L. F. Vecchietti**, D. Han, D. Li, S. W. Lee, "**Mitigating Plasticity Loss through Architectural Design in Continual Learning**," in International Conference on Machine Learning (ICML), 2026.

[3] J. Kim, J. Kwon, **L. F. Vecchietti**, W. Dong, J. Kim, M. Cha, "**Machine Behavior in Relational Moral Dilemmas: Moral Rightness, Predicted Human Behavior, and Model Decisions**," in ACL Findings, 2026.

[4] **L. F. Vecchietti**, M. Lee, B. Hangeldiyev, B. N. Wijaya, H. Jung, H. Park, T-K Kim, M. Cha, H. M. Kim, "**Interpretable Machine Learning for Protein Science: Structure, Function, and Interactions**," in ACM Computing Surveys, 2026.

[5] J. Kwon, **L. F. Vecchietti**, S. Park, M. Cha, "**Dropouts in Confidence: Moral Uncertainty in Human-LLM Alignment**," in AAAI, AI Alignment Track, 2026.

[6] **L. F. Vecchietti**, B. N. Wijaya, et al., "**Artificial intelligence-driven computational methods for antibody design and optimization**," mAbs, 2025, doi: 10.1080/19420862.2025.2528902.

[7] J. Yun, S. Yang, J. H. Kwon, **L. F. Vecchietti**, M. Cha, J. E. Oh, H. M. Kim, "**Computational Design and Glycoengineering of Interferon-Lambda for Nasal Prophylaxis against Respiratory Viruses**," Advanced Science, 2025.

[8] M. Lee, **L. F. Vecchietti** *(co-first author)*, H. Jung, H. J. Ro, M. Cha, H. M. Kim, "**Robust Optimization in Protein Fitness Landscapes Using Reinforcement Learning in Latent Space**," in International Conference on Machine Learning (ICML), 2024 *(Spotlight Poster)*.

[9] M. Seo, **L. F. Vecchietti**, S. Lee, D. Har, "**Rewards Prediction-Based Credit Assignment for Reinforcement Learning With Sparse Binary Rewards**," in IEEE Access, vol. 7, pp. 118776-118791, 2019, doi: 10.1109/ACCESS.2019.2936863.

[10] T. B. Ribeiro, **L. F. Vecchietti**, et al., "**Overabundance of abelisaurid teeth in the Acu Formation (Albian-Cenomanian), Potiguar Basin, Northeastern Brazil: morphometric, cladistic and machine learning approaches**," Journal of Vertebrate Paleontology, 2025.

[11] S. Mishra, P. K. Rajendran, **L. F. Vecchietti**, D. Har, "**Sensing accident-prone features in urban scenes for proactive driving and accident prevention**", 2023, IEEE Transactions on Intelligent Transportation Systems.

[12] **L. F. Vecchietti**, M. Seo, D. Har, "**Sampling Rate Decay in Hindsight Experience Replay for Robot Control**," in IEEE Transactions on Cybernetics, 2020, doi: 10.1109/TCYB.2020.2990722.

[13] **L. F. Vecchietti**, T. Kim, K. Choi, J. Hong, D. Har, "**Batch Prioritization in Multigoal Reinforcement Learning**," in IEEE Access, vol. 8, pp. 137449-137461, 2020, doi: 10.1109/ACCESS.2020.3012204.

[14] C. Hong, I. Jeong, **L. F. Vecchietti**, D. Har, J. -H. Kim, "**AI World Cup: Robot Soccer-Based Competitions**," in IEEE Transactions on Games, 2021, doi: 10.1109/TG.2021.3065410.

[15] T. Kim, **L. F. Vecchietti**, K. Choi, S. Sariel, D. Har, "**Two-stage training algorithm for AI robot soccer**", in PeerJ Computer Science, 7:e718, 2021, https://doi.org/10.7717/peerj-cs.718.

[16] S. Lee, **L. F. Vecchietti**, H. Jin, J. Hong, D. Har, "**Power Management by LSTM Network for Nanogrids**," in IEEE Access, vol. 8, pp. 24081-24097, 2020, doi: 10.1109/ACCESS.2020.2969460.

[17] S. Lee, H. Jin, **L. F. Vecchietti**, J. Hong, D. Har, "**Short-Term Predictive Power Management of PV-Powered Nanogrids**," in IEEE Access, vol. 8, pp. 147839-147857, 2020, doi: 10.1109/ACCESS.2020.3015243.

[18] T. Kim, **L. F. Vecchietti**, K. Choi, S. Lee, D. Har, "**Machine Learning for Advanced Wireless Sensor Networks: A Review**," in IEEE Sensors Journal, vol. 21, no. 11, pp. 12379-12397, 2021, doi: 10.1109/JSEN.2020.3035846.

[19] S. Kim, I. Kim, **L. F. Vecchietti**, D. Har, "**Pose Estimation Utilizing a Gated Recurrent Unit Network for Visual Localization**", in Applied Sciences, 10, no. 24: 8876, 2020, https://doi.org/10.3390/app10248876.

[20] S. Lee, D. Har, **L. F. Vecchietti**, J. Hong, H. -J. Lim, "**Optimal Link Scheduling Based on Attributes of Nodes in 6TiSCH Wireless Networks**", in The Journal of Korean Institute of Information Technology, vol. 18, no. 1, pp.77-92, 2020, https://doi.org/10.14801/jkiit.2020.18.1.77.

[21] S. Lee, H. Jin, **L. F. Vecchietti**, J. Hong, K-B. Park, PN. Son, D. Har, "**Cooperative decentralized peer-to-peer electricity trading of nanogrid clusters based on predictions of load demand and PV power generation using a gated recurrent unit model**", in IET Renewable Power Generation, vol. 15, pp. 3505-3523, 2021, https://doi.org/10.1049/rpg2.12195.

Preprints

[1] L. Vecgaile, A. Spata, **L. F. Vecchietti**, E. Zagheni, "**Predicting Individual Life Trajectories: Addressing Uncertainty in Social Employment Transitions**," SocArXiv, 2025.

Workshops, Domestic Conferences

[1] B. N. Wijaya, **L. F. Vecchietti**, A. Cho, C. Cha, M. Cha "**Longitudinal Identification of Critical Factors in Nurse Turnover for Supporting Workforce Well-Being**", in HCI Korea, 2026.

[2] J. Kim, J. Kwon, **L. F. Vecchietti**, A. Oh, M. Cha, "**Exploring Persona-dependent LLM Alignment for the Moral Machine Experiment**," arXiv preprint arXiv:2504.10886 (2025), presented at the ICLR 2025 BiAlign Workshop.

[3] B. N. Wijaya, **L. F. Vecchietti**, M. Cha, H. M. Kim, "**Evaluating Antibody Structure Reconstruction with an SE(3)-Equivariant Graph Neural Network**", in Proceedings of the Korea Software Congress (KSC), 2023 *(Excellence Award)*.

[4] M. Lee, **L. F. Vecchietti**, H. Jung, M. Cha, H. M. Kim, "**Protein Sequence Design in a Latent Space via Model-based Reinforcement Learning**", 2022, presented at the NeurIPS 2022 Machine Learning in Structural Biology Workshop.

[5] M. Lee, A. Rzayev, H. Jung, **L. F. Vecchietti**, M. Cha, H. M. Kim, "**Structure-based representation for protein functionality prediction using machine learning**", in Proceedings of the Korea Computer Congress (KCC), 2022 *(Excellence Award)*.

[6] P. K. Rajendran, S. Mishra, **L. F. Vecchietti**, D. Har, "**RelMobNet: End-to-end relative camera pose estimation using a robust two-stage training**", 2022, arXiv preprint arXiv:2202.12838, presented at the ECCV 2022 IWDSC Workshop.

[7] B. Hangeldiyev, A. Rzayev, A. Armanuly, **L. F. Vecchietti**, M. Cha, H. M. Kim, "**Antibody Sequence Design With Graph-Based Deep Learning Methods**", in Proceedings of the Korea Software Congress (KSC), 2022.

Work Experiences

Hyundai Motor Company, Namyang Research and Development Center *Namyang, South Korea, Jul. 2013*

Intern. Advisor: Dong-pil Yoon.

Honors and Awards

Science without Borders Program *Feb. 2013 - Feb. 2014*

Brazilian government 1-year scholarship for undergraduate students with outstanding academic achievements.

KAIST Scholarship *Aug. 2017 - Feb. 2021*

Ph.D. Scholarship.

Academic Services

Reviewer

Journals: IEEE Transactions on Cybernetics, IEEE Sensors, IEEE Transactions on Games, Frontiers in AI and Robotics, ACM Transactions on Intelligent Systems and Technology

Conferences: AAAI ICWSM 2022, NeurIPS 2024-2026, ICLR 2025-2026, WSDM 2025, AISTATS 2025, ICML 2025-2026, AAAI 2026, Machine Behavior (Ma+Be) Conference 2026

Workshops: ICML LatinX 2021, ICLR Reincarnating RL 2023, NeurIPS MLSB 2023-2025, ICML ML4LMS 2023, ICML MALGAI 2026

Invited Presentations

Center for Neuroscience-inspired AI - KAIST, South Korea *October 2025*

Invited to present a lecture titled "*From self-organized networks to deep reinforcement learning: perspectives on AI research.*"

Ruhr University Bochum, Germany *July 2025*

Invited to present a lecture titled "*LLMs outside Natural Language Processing applications.*"

WebImmunization Seminar, University of Oslo, Norway *December 2024*

Invited to present a talk titled "*Integrating Data Science and AI methods in multidisciplinary research to make discoveries with social impact.*"

Cradle Bio, Zurich, Switzerland *November 2024*

Invited to present a talk titled "*Robust Optimization in Protein Fitness Landscapes Using Reinforcement Learning in Latent Space.*"

Graduate School of AI, Gwangju Institute of Science and Technology (GIST), South Korea *July 2023*

Invited to present a talk titled "*Developing and applying deep learning methods for protein design*."

Max Planck Institute for Security and Privacy (MPI-SP), Germany *May 2023*

Invited to present a talk titled "*Developing and applying deep learning methods to facilitate new scientific discoveries.*"

IBS Winter School on AI-Boosted Basic Science - Institute for Basic Science (IBS), South Korea *Dec. 2022*

Invited to present a talk titled "*Target-conditioned protein and antibody design for drug discovery.*"

School of AI Convergence - Chonnam National University, South Korea *Nov. 2021*

Invited to present a work titled "*Identifying the key actions that lead an agent to accomplish a task in model-based deep reinforcement learning*."

Cho Chun Shik Graduate School of Green Transportation - KAIST, South Korea *Oct. 2021*

Invited to present a work titled "*Performance enhancement in multigoal model-based deep reinforcement learning*."

Institute for Basic Science (IBS), South Korea *Apr. 2021*

Invited to present a work titled "*Identifying the key actions that lead an agent to accomplish a task in model-based deep reinforcement learning*."

Open-Source Systems

Dino Toothfier (lead developer) [Code]

Machine learning-based application to classify Dinosaur teeth.

AI Soccer Robot Simulator (lead developer) [Code]

Simulator to train multi-agent AI algorithms for robot soccer.